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Study of the performance of different subpixel image correlation methods in 3D digital image correlation

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The three-dimensional digital image correlation (3D-DIC) method is rapidly developing and is being widely applied to engineering and manufacturing. Despite its extensive use, the error caused by different image matching algorithms is seldom discussed. An algorithm for 3D speckle image generation is proposed, and the performances of different subpixel correlation algorithms are studied. The advantage is that there is no interpolation bias of texture in the simulation before and after deformation, and the error from the interpolation of speckle can be omitted in this algorithm. An error criterion for 3D reconstruction is proposed. 3D speckle images were simulated, and the performance of four subpixel algorithms is addressed. Based on the research results of different subpixel algorithms, a first-order Newton–Raphson iteration method and gradient-based method are recommended for 3D-DIC measurement. © 2010 Optical Society of America

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1. Introduction

Three-dimensional digital image correlation (3D-DIC), using stereovision in conjunction with DIC, is a full-field, robust, and convenient method used for shape reconstruction and deformation analysis [1]. It has been proven to be a powerful noncontact technique for measuring the displacement or strain field of any 3D object. It is currently used prevalently in materials science, engineering mechanics [2], biomechanics [3], polymer forming [4], high-speed impact testing [5,6], and many other fields.

The accuracy of 3D reconstruction depends on many factors. There are two viewpoints on error sources. From the first viewpoint [7,8], there are two error sources. One is hardware and layout, including the quality of the cameras and their resolution, and the configuration of the two cameras (distance between the two cameras, angle between their optical axis, etc.), which governs the triangulation accuracy. The other source is the algorithm, the stereovision sensor calibration and the correlation.

The other viewpoint is from Siebert, who conducted a series of studies of the error estimation in 3D-DIC techniques [9–11]. The error sources can be divided into two categories according to their effect on the evaluation process: one error source is "correlation error," which influences the accuracy of the image correlation. The other category is calibration errors, which have a direct effect on the reconstruction of the 3D coordinates of correlated image facets, also called "reconstruction errors" [11].

Although the error sources of 3D-DIC are discussed in the literature [1,7–11], the performance of different image correlation algorithms in 3D-DIC has not been taken into consideration. In [12], the authors investigated three types of the most commonly used subpixel displacement registration algorithms to study their effect on 2D-DIC deformation in terms of the registration accuracy and the computational efficiency using computer-simulated speckle images. However, the correlation procedure of 2D-DIC, which is based mainly on the plane assumption,

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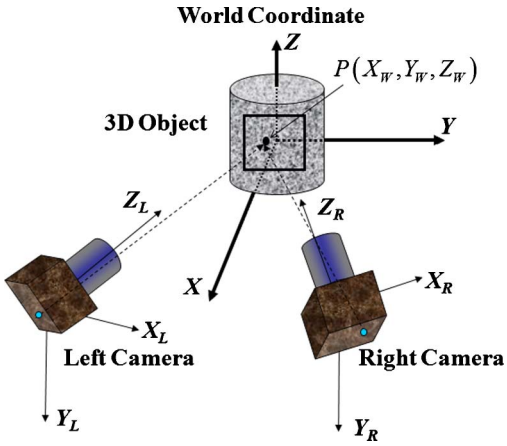

Fig. 1. (Color online) Schematic of 3D-DIC system.

is quite different from 3D-DIC. Schreier [13] studied the accuracy of 3D image correlation by using projective constraints and unconstrained correlation and concluded that using unconstrained correlation is more accurate in 3D image correlation. Thus, in this paper, the 3D image correlation methods using unconstrained correlation are employed.

In the 2D-DIC method, the subpixel correlation algorithms are commonly used. These subpixel algorithms include Newton–Raphson (NR) iteration [14–16], intensity interpolation (i.e., the coarse–fine search method [17,18]), correlation coefficient curve fitting or interpolation [19,20], the double Fourier transform [21,22], gradient-based methods [15,23,24], the artificial neural network method [25,26], and genetic algorithm [27,28]. Among them, the NR iteration, correlation coefficient curve-fitting method, and gradient-based method are the three most commonly used algorithms in most reference literature owing to their simplicity, effectiveness, and accuracy [12]. Without doubt, these subpixel algorithms can be extended to be used in 3D-DIC correlation. Because of the single-plane assumption in 2D-DIC, the effect of

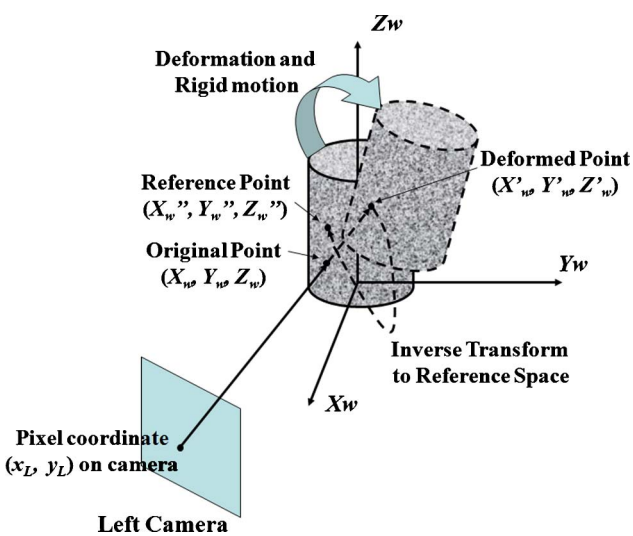

Fig. 2. (Color online) Procedure of 3D speckle image generation.

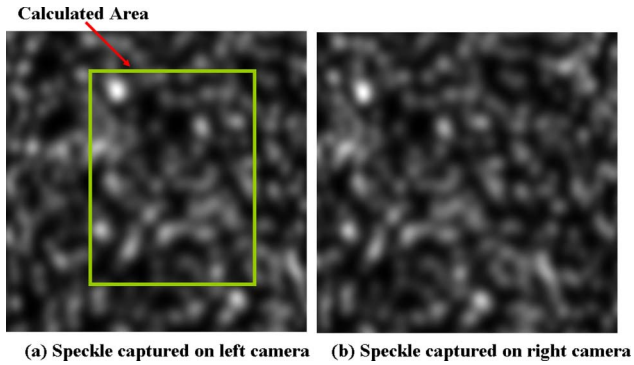

Fig. 3. (Color online) Pair of images captured by left- and right-hand cameras.

2D subpixel algorithms on the accuracy of 3D-DIC is still unknown. Naturally, knowing how to choose an appropriate subpixel algorithm in 3D-DIC may be difficult. Therefore, understanding the limitations and performance of these subpixel registration algorithms in 3D image correlation is necessary and also very helpful.

An algorithm for 3D speckle image generation is proposed in this research. This algorithm has high accuracy and omits the error source of interpolation of textures onto 3D objects. Based on the computer-generated 3D speckle images, the performance accuracy of four types of commonly used subpixel registration algorithm for 3D shape reconstruction and deformation was studied.

The outline of this paper is as follows: the principles of 3D-DIC and the accuracy evaluation standard are introduced in Section 2. The 3D speckle image generation algorithm is described in Section 3. Four types of subpixel correlation algorithm are briefly presented in Section 4, and the results of different types of 3D image correlation are discussed.

2. Principle and Error Criterion of 3D-DIC

Binocular stereovision is a technique for recovering the 3D structure of a scene from two different viewpoints. The principle of 3D-DIC was first proposed by Luo *et al.* [29] and has presented in many studies [1,30–32]. The optical system is shown in Fig. 1. Here, a brief description is presented.

Point $P(X_W, Y_W, Z_W, 1)$ is a 3D point in world coordinates expressed as homogenous coordinates. Suppose that the rotation matrix and translation vector relative to the world coordinate system are R and T, respectively, with the camera parameter K. Thus the pixel $p = [x \quad y \quad 1]^T$ on the camera sensor and the world point P have the relationship (provided that the camera satisfies a pinhole model)

$$\lambda p = K \begin{bmatrix} R & T \\ 0 & 1 \end{bmatrix}^{-1} P, \tag{1}$$

where λ is a ratio scale.

At least two cameras are required in order to determine the world coordinate point. As a result,

Table 1. Parameters of 3D Speckle Image Generation

L	$K_L = \begin{bmatrix} 4000 & 0 & 256 \\ 0 & 4000 & 256 \\ 0 & 0 & 1 \end{bmatrix}$	$R_L = \begin{bmatrix} 0.1305 & 0 & -0.9914 \\ 0.9914 & 0 & 0.1305 \\ 0 & -1 & 0 \end{bmatrix}$	$T_L = \begin{bmatrix} 396.5579 \\ -52.2105 \\ 0 \end{bmatrix}$
R	$K_R = \begin{bmatrix} 4000 & 0 & 256 \\ 0 & 4000 & 256 \\ 0 & 0 & 1 \end{bmatrix}$	$R_R = \begin{bmatrix} -0.1305 & 0 & -0.9914 \\ 0.9914 & 0 & -0.1305 \\ 0 & -1 & 0 \end{bmatrix}$	$T_R = \begin{bmatrix} 396.5579 \\ 52.2105 \\ 0 \end{bmatrix}$

the 3D-DIC technique generally utilizes two cameras to simultaneously capture the same regions of interest on the object surface from two different viewpoints. To determine the 3D coordinates of point P, the first step is camera calibration, followed by image correlation, and finally the triangulation method.

In order to study the accuracy of different image correlation methods, an error criterion has been proposed. Based on Eq. 9 of [7], a more physically meaningful criterion is suggested:

$$J = \frac{\sum_{i=1}^{N} \sqrt{(\Delta X_{wi}^2 + \Delta Y_{wi}^2 + \Delta Z_{wi}^2)}}{N}, \tag{2}$$

where N is the sum of all points in the effective field of view, $\Delta X_{wi}, \Delta Y_{wi}, \Delta Z_{wi}$ are the ith absolute errors of reconstructed coordinates relative to the original, and J is the average distance from the original 3D point.

3. Introduction of 3D Speckle Image Generation

In [13], virtual test bed (VTB) software for 2D and 3D-DIC analysis was designed and implemented. The VTB can be divided into two parts: a modular, object-oriented rendering engine and a user interface layer implemented as interpreted scripting language. The VTB has two main challenges in the design of the ray tracer. One is the accurate mapping of textures, e.g. speckle patterns, onto deforming surfaces, while the other is the accurate interpolation of textures.

A much more convenient method called 3D speckle image generation is proposed. The proposed method does not need interpolation before and after deformation of texture. The texture on the object is accurate, based on the theoretical derivation.

The procedure of 3D speckle image generation is illustrated in Fig. 2 and is presented as follows:

a. Set the extrinsic and intrinsic camera parameters correctly. To conveniently describe this method, the parameters of the left-hand camera are taken as illustration. The rotation matrix is R_L, and the translation vector is T_L. Assume that the object has a surface function expressed as

$$\xi(X_W, Y_W, Z_W) = 0, \tag{3}$$

where X_W, Y_W, and Z_W are the world coordinates.

b. Note that a ray for every pixel in the camera sensor has a intersection point with the object surface according to the imaging characteristics expression

$$\lambda_L \underset{\sim}{x_L} = \underset{\sim}{P_L} \underset{\sim}{X_W}, \tag{4}$$

where $x_L = [x_L \quad y_L \quad 1]^T$ is the homogeneous coordinate of the camera pixel and

$$P_L = K_L \begin{bmatrix} R_L & T_L \\ 0 & 1 \end{bmatrix}^{-1}$$

is the homograph matrix. Based on Eqs. (3) and (4), the intersection point on the object surface could be calculated, shown in Fig. 2 as the "original point."

c. After world coordinate calculation, and similar to 2D speckle image generation [23], speckle patterns on the object surface before and after deformation are assumed to be the sum of individual Gaussian speckles:

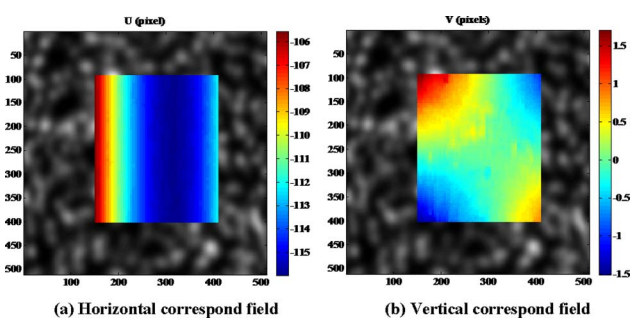

Fig. 4. (Color online) Correlation result using first-order NR iteration method.

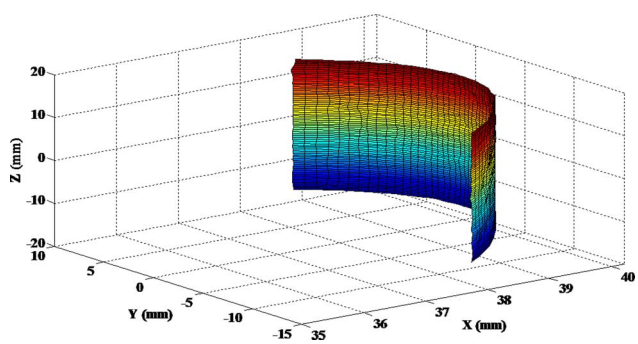

Fig. 5. (Color online) 3D shape reconstruction of the cylinder with radius 40 mm.

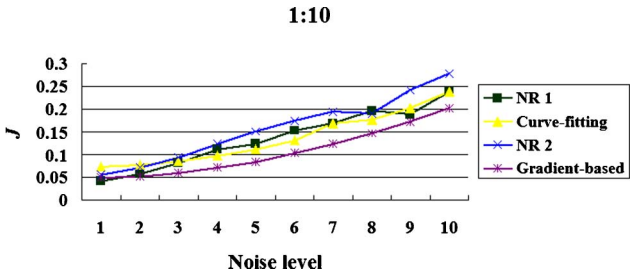

Fig. 6. (Color online) Left-hand noise level 1 correlation to ten right-hand images.

$$I(X_W, Y_W, Z_W) = \sum_{k=1}^{s} I_k^0 \times \exp\left[-\frac{(X_W - X_W^k)^2 + (Y_W - Y_W^k)^2 + (Z_W - Z_W^k)^2}{R_0^2}\right], \tag{5}$$

where s is the total number of speckles, R_0 is the speckle size, (X_W^k, Y_W^k, Z_W^k) are positions of each speckle with a random distribution, and I_k^0 is the peak intensity of each speckle. After substituting the original point coordinates into Eq. (5), the reference image can be generated.

d. After deformation, the surface of the object has the expression

$$\xi(X'_W, Y'_W, Z'_W) = 0. \tag{6}$$

From Eqs. (4) and (6), the "deformed point" (X'_W, Y'_W, Z'_W), as illustrated in Fig. 2, can be calculated. Also the deformed point could be transformed to the reference space as reference point (X''_W, Y''_W, Z''_W) by implementing the continuum mechanics solution of deformed object ("reference point," Fig. 2). After substituting the reference point coordinates into Eq. (5), the deformed speckle images can be generated.

4. Performance of Different Image Correlation Algorithms

In this section, not only the accuracy of shape reconstruction, but also the accuracy of deformation caused by image correlation algorithm is analyzed.

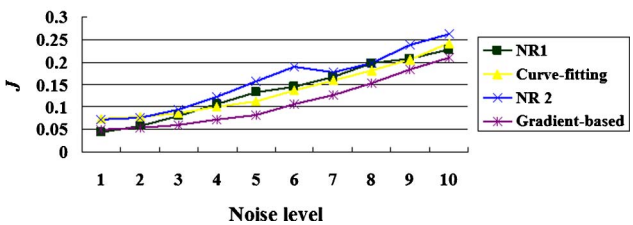

Fig. 7. (Color online) Left-hand noise level 2 correlation to ten right-hand images.

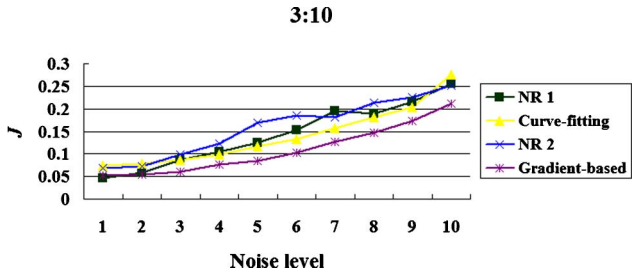

Fig. 8. (Color online) Left-hand noise level 3 correlation to ten right-hand images.

A. Introduction of Image Correlation Algorithms

The premise of an image correlation measurement technique is that a point on an object's surface does not change its relative intensity as the object undergoes deformation. The basic principle is to match two speckle patterns between two images. Typically, a subset of $(2M+1)\times(2M+1)$ pixels from the left-hand image is chosen to find its location in the right-hand image. For the best estimation of the displacements based on situation of 3D setup, the following zero-normalized sum of squared differences coefficient is most commonly used [33]:

$$C_{f,g}(\vec{h}) = \sum_{x=-M}^{M}\sum_{y=-M}^{M}\left[\frac{f(x,y)-f_m}{\sqrt{\sum_{x=-M}^{M}\sum_{y=-M}^{M}[f(x,y)-f_m]^2}} - \frac{g(x+\xi,y+\eta)-g_m}{\sqrt{\sum_{x=-M}^{M}\sum_{y=-M}^{M}[g(x+\xi,y+\eta)-g_m]^2}}\right]^2, \tag{7}$$

where $f(x,y)$ and $g(\xi,\eta)$ are the gray values of the subset centered at the source and the target points located in the reference and the deformed images, respectively; ξ, η are so-called shape functions or displacement mapping functions. The influence of illumination lighting fluctuation can be alleviated by using proper correlation criterion [34]. The zero-normalized sum of squared differences coefficient is recommended for correlation analysis, which is insensitive to the offset and linear change of illumination lighting [33].

Four classic subpixel algorithms are used for the study of image correlation performance, including first-order NR iteration [14], second-order NR

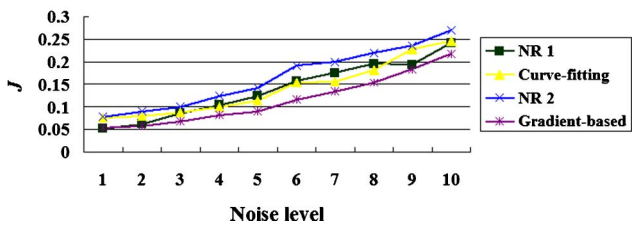

Fig. 9. (Color online) Left-hand noise level 4 correlation to ten right-hand images.

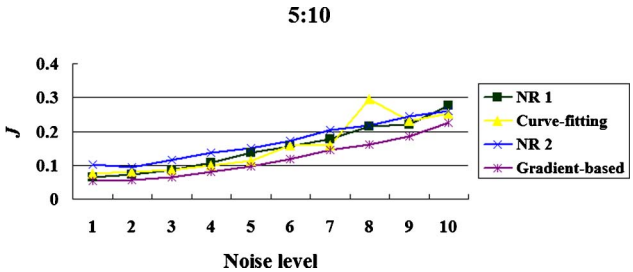

Fig. 10. (Color online) Left-hand noise level 5 correlation to ten right-hand images.

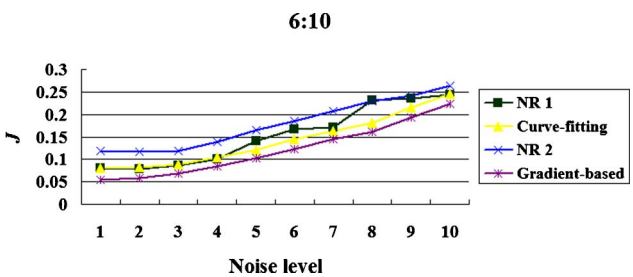

Fig. 11. (Color online) Left-hand noise level 6 correlation to ten right-hand images.

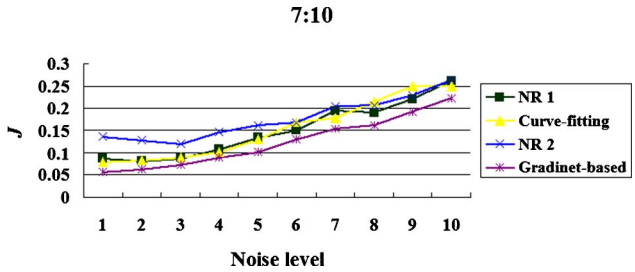

Fig. 12. (Color online) Left-hand noise level 7 correlation to ten right-hand images.

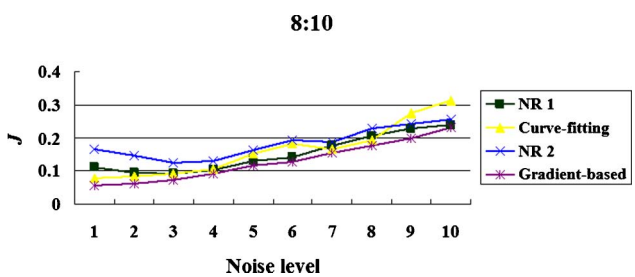

Fig. 13. (Color online) Left-hand noise level 8 correlation to ten right-hand images.

iteration [16], the curve-fitting algorithm [12], and the gradient-based method [23]. The NR method takes the deformation of the concerned subset between the two images into account. After the shape function is substituted into Eq. (7), the nonlinear equation can be solved by using the NR iteration method (or the Levenberg–Marquart algorithm). The integer pixel search result can be used as an initial iteration guess. When a second-order shape function is substituted into the coefficient function, it becomes the second-order NR algorithm. The curve-fitting method selects its neighboring pixel locations to constitute a surface that can be represented by a 2D function. Thus the position of the maximum value of the coefficient can be located. The gradient-based method was first developed by Davis and Freeman [15] as an optical flow method. Based on the assumption of the gradient-based subpixel registration DIC, when the subset is small enough, the subset rigid body translation exists. Reliability-guided DIC [35] analysis begins with a selected seed point with the highest correlation coefficient and proceeds with neighbors around that point. It is an effective correlation procedure. This method is employed in our research software.

B. Image Simulation

To study the performance of different image correlation algorithms, a 3D object, which is a cylinder with 40 mm radius, was set up in the world coordinates as shown in Fig. 1. The speckle images for the left and right cameras are simulated as shown in Fig. 3. The parameters are shown in Table 1, where the subscripts L and R indicate the left and right camera parameters, respectively, K is the intrinsic parameter, and R and T are, respectively, the rotation matrix and translation vector relative to the world coordinate system.

C. Results and Analysis

Figure 3 is a pair of ideal images. After the image correlation procedure for the rectangular area in Fig. 3, the correlated horizontal and vertical results are shown in Figs. 4(a) and 4(b), respectively. Figure 5 shows the reconstructed 3D shape of the cylinder.

1. *Performance in Accuracy of Reconstruction Shape*

In practical experiments, the camera captured image is usually affected by different noise sources such as the illumination, the camera, and environmental vibration. Thus random Gaussian noise was added to the left and right images; the noise was classified into ten levels with mean zero and standard deviation from 0.5 to 5 with step 0.5. Figure 6 shows the results of the left-hand noise level 1 image correlated with the right-hand for ten noise level images, obtained by using these four correlation algorithms, where the horizontal axis is different noise levels and the vertical axis is error criterion J, where NR 1 and NR 2 are the first-order NR and the second-order NR method, respectively. Figures 6–15 illustrate the entire left-hand different level noise images correlated with the ten right-hand images obtained by the four correlation methods.

Figure 16 illustrates the 3D object shape reconstruction from the top view of Z, and the correlation images are one left-hand noise level image correlated with the same noise level of the right-hand image. From the images, the result of the first-order NR method is much smoother than the others, especially

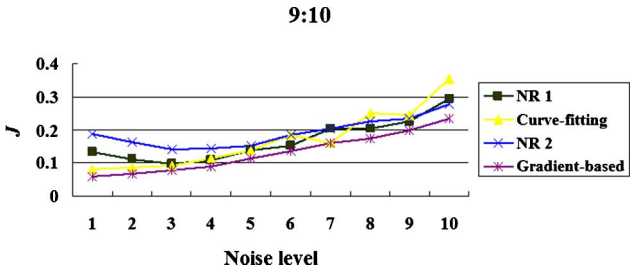

Fig. 14. (Color online) Left-hand noise level 9 correlation to ten right-hand images.

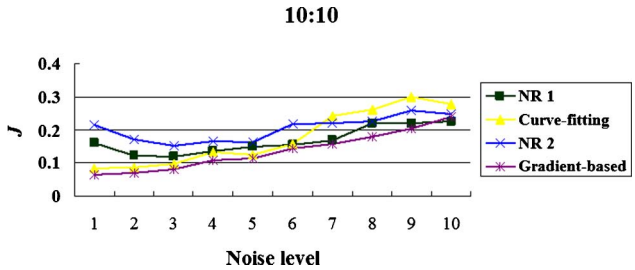

Fig. 15. (Color online) Left-hand noise level 10 correlation to ten right-hand images.

compared with the most time-consuming method, the second-order NR method. The results of the curve-fitting method and the second-order NR method are rough, and the edge of gradient-based method's reconstructed shape is rougher than the middle area.

From these results, it can be seen that in most cases the gradient-based method is the best compared with the other three algorithms because it has the smallest J. Also, results of the curve-fitting method have a smaller J than that of second-order NR, which is quite different from the 2D-DIC performance [12]. This is understandable because 2D-DIC is based on the plane assumption, and the optical axis of the camera is normal to the surface of the objects, which is unlike in 3D-DIC. In most cases, the first-order NR method is more insensitive to the noise level than curve fitting. Also, more general simulated 3D speckle images are studied with different radii. The results pooled from these simulations exhibit lead to almost the same conclusion, with the main difference being that the results of the gradient-based method are dispersed according to the calculated area. From Fig. 16, we can see that the gradient-based method has the larger error at the edge of the image cylinder. To compare the method's computational efficiencies, Table 2 illustrates their time consumption relative to that for the curve-fitting method.

From these results, we can conclude that when two images are correlated, the image with the least amount of noise should be used as the object image, and the higher-noise image should be the reference image, because the interpolation is performed on the object image subset field. From a comparison of the accuracy and time consumption, the gradient-based method is a good choice for 3D shape

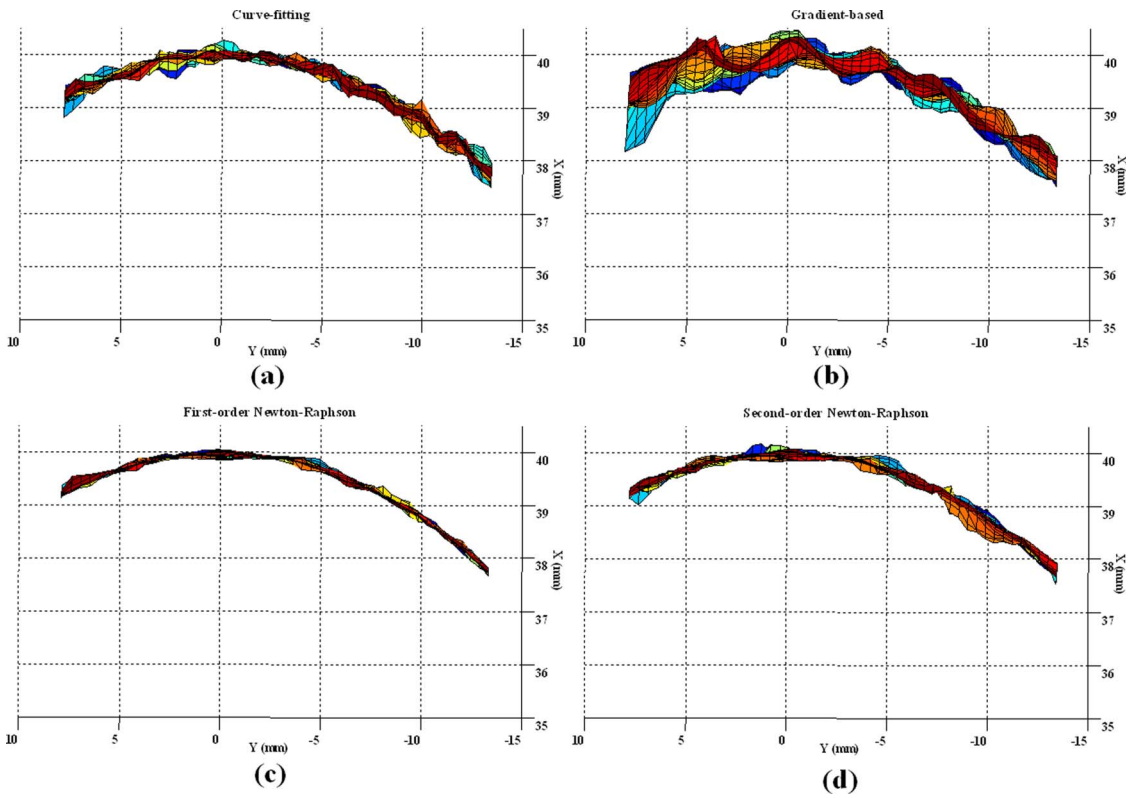

Fig. 16. (Color online) 3D shape reconstructed by using (a) first-order NR, (b) coefficient fitting, (c) second-order NR method, (d) gradient-based method.

Table 2. Relative Time Consumption

Algorithm	Relative Time, 29 × 29 Pixels Subset
First-order NR method	8.96
Curve-fitting method	1
Second-order NR method	13.2
Gradient-based method	1.89

reconstruction in 3D-DIC, but the calculated area should be carefully chosen.

2. *Performance of Accuracy of Deformation*

A series of simulated 3D speckle deformation images were investigated. Assume that the cylinder object was translated along X-axis direction (out-of plane) for 0.01–0.1 mm with step of 0.01 mm and for 0.1–1 mm with a step of 0.1 mm. The four algorithms were applied for the deformation analysis. The standard deviation (SD) of deformation calculated was obtained as shown in Fig. 17. The SD can be regarded as systematic error, shown as the Y axis. The X axis is the base 10 logarithm of the deformation for clear illustration as shown in Fig. 17.

From this figure, we can conclude that the gradient-based method and the first-order NR method have the same error level and that both are very sensitive to small deformation, while the latter is smoother. In Fig. 17, the second-order NR method and curve-fitting method show the same distribution, and the curve-fitting has the highest systematic error in most cases. An interesting phenomenon is that the SD error of these four methods at displacements of 0.04 and 0.4 mm decreases suddenly. One possible explanation could be that the speckle images have multiples of half or integer pixel movement before and after object movement.

In summary, the gradient-based method and the first-order NR method are preferred to the other two algorithms because of the smaller systematic error and better sensitivity. Also, when comparing the time involved, we can see that the gradient-based algorithm is preferable. More simulation results demonstrated that the first-order NR method is more insensitive to the calculated area than the gradient-based method. However, the calculated area should be carefully chosen when implementing the gradient-based method. In further research, simulations of two cameras with different intrinsic and extrinsic parameters were studied, and the conclusion was still valid according to our simulation: the gradient-based method is preferable because of its advantage in time consumption and accuracy.

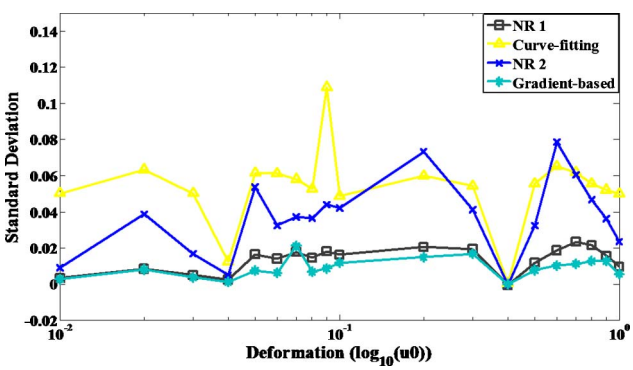

Fig. 17. (Color online) Systematic errors of four subpixel registration algorithm for rigid 3D object translation (using a 29 × 29 pixel subset).

5. Conclusion

In this paper, an accurate algorithm called 3D speckle image generation has been proposed. The procedure of 3D speckle image generation could omit the error sources from interpolation of texture and bias of intensity. The proposed 3D speckle image generation could be used for further research in numerical analysis in 3D-DIC.

Among the four algorithms compared, the gradient-based method has the best accuracy both in cylinder 3D shape reconstruction and deformation measurement, but it is sensitive to calculated area. This conclusion is still valid when the two cameras have different intrinsic and extrinsic parameters according to our simulation; however, this case needs further study. Based on the results of different noise level images for object shape reconstruction, if two images do not have the same noise level, in image correlation in 3D-DIC the higher-noise image should be regarded as the reference image, which can improve the accuracy of 3D object shape reconstruction. From these results of cylinder reconstruction and deformation, the first-order NR method is still preferable in 3D-DIC, just as in 2D-DIC, compared with the curve-fitting method. A high quadratic shape such as the second-order function does not effectively improve the accuracy of cylinder shape reconstruction and deformation measurement in 3D-DIC. Whether the conclusion can be effectively applied to real experimental images contaminated with a variety of complicated noises or with different 3D object shape reconstructions still requires further research. In spite of this, the four algorithms applied in practical experimental images always exhibit the same performances demonstrated in this research.

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